

STAGE & STEEL

HEADLESS D2C SPORTS NUTRITION PLATFORM — COMPLETE
TECHNICAL AUDIT & COMMERCIAL PROPOSAL

PRODUCTION
BUILD V1.0

Built with Next.js 16 +
React 19

ARCHITECTURE

Headless D2C
Stack

LOGISTICS ENGINE

Delhivery Direct
API

PAYMENT ENGINE

Cashfree Seamless
SDK

DATABASE & AUTH

Google Firebase /
Cloud

Executive Overview: This platform is a fully custom, enterprise-grade headless D2C e-commerce engine engineered for ultra-high conversion, instant loading speeds (<1s), automated shipping fulfillment, and luxury athletic brand aesthetics. Unlike template-based solutions (Shopify/WooCommerce), this system operates with **zero recurring monthly app subscription costs** and delivers proprietary 3D/Canvas simulations and sound engineering.

01 VISUAL EXPERIENCE & CUSTOM ANIMATION ENGINE

HIGH-VALUE INTERACTIVE UI/UX ELEMENTS

Custom 2D/3D Canvas & Web Audio

- **Custom 2D/3D Canvas Powder Simulator (PowderParticleCanvas.tsx):** Real-time physics engine rendering realistic supplement micro-dust and billowing chalk smoke with particle velocity, air drag resistance, radial gradients, and smooth alpha decay.
- **Interactive Supplement Scoop Animation (SupplementScoop.tsx):** 3D tilting scoop experience triggering real-time interactive powder pouring upon user engagement and cart additions.
- **Ambient Floating Gym Dust (AmbientDust.tsx):** Atmospheric micro-particle background creating an authentic hardcore athletic visual aura.
- **Custom Web Audio API Sound Synthesizer (sound.ts):** Zero-latency mathematical audio oscillators generating tactile mechanical clicks and dual-tone harmonic "Add-to-Cart" chimes without heavy external audio file loading.
- **Interactive Powder Mega Menu (PowderMegaMenu.tsx):** Multi-level interactive navigation menu with live visual category transitions and instant catalog filtering.
- **Brutalist Typography & Luxury Dark Theme:** Integrated Google Fonts (*Oswald*, *Barlow Condensed*, *Space Grotesk*, *Inter*) styled with military-grade olive accents.

02 PRODUCT CATALOG & LAB-TESTED TRANSPARENCY SUITE

CONVERSION-DRIVEN PRODUCT ARCHITECTURE

Trust & Lab COA Engine

- **Dynamic Category Matrix:** Real-time category filtering across *All Products*, *Protein Matrix*, *Creatine Lab*, and *EAA Aminos* with direct URL hash synchronization (#protein, #creatine, #eaa).
- **Interactive Product Deep-Dive Modal (ProductModal.tsx):** Multi-angle high-resolution gallery (Front pack, Supplement facts, Amino profile & Directions).
- **Official Lab Test COA Viewer:** Embedded authentic Certificate of Analysis (COA) lab reports directly in product view to build unbreakable consumer trust.
- **Per-Scoop Nutritional Breakdown:** Batch code verification badge, 25g protein/energy matrix stats, and live flavor switching with instant stock/price updates.

03 AUTHENTICATION, SECURITY & USER ACCOUNTS

FIREBASE CLOUD AUTHENTICATION (AUTHCONTEXT.TSX, AUTHMODAL.TSX)

Social + Email Auth

- **Google 1-Tap & OAuth Popup Login:** Instant, frictionless social sign-in.
- **Secure Email/Password Authentication:** Encrypted user registration and credential management.
- **Frictionless Guest Checkout Mode:** Customers can complete transactions in seconds without mandatory sign-up.
- **State Synchronization:** Persistent session management across page refreshes and devices.

04 SMART CART DRAWER & DYNAMIC COUPON ENGINE

REVENUE OPTIMIZATION & PROMOTION MANAGEMENT

Dynamic Discount Rules

- **Slide-Out Cart Drawer (CartDrawer.tsx):** Smooth side-drawer cart with live quantity modification, instant subtotal recalculations, and savings indicators.
- **Dual-Mode Discount Engine (lib/coupons.ts):**
 - **Percentage Discounts:** Code-based discounts (LAUNCH10, STAGE10, DIVESH10, ASHISH10) with customizable discount caps.
 - **Flat Discounts:** Minimum order threshold discounts (e.g., LAUNCH200 for orders above ₹1,500).
- **Instant Client Validation:** Real-time feedback banners with clear savings notifications.

05 CASHFREE PAYMENT GATEWAY INTEGRATION

SEAMLESS IN-APP PAYMENTS (CHECKOUTMODAL.TSX, /API/CASHFREE/*)

Zero Redirection Drops

- **Server-to-Server Session Generation:** Secure REST API creating cryptographic payment sessions.
- **In-App Modal Checkout:** High-converting modern modal supporting UPI (GPay, PhonePe, Paytm), Credit/Debit Cards, NetBanking, and Wallets without leaving the domain.
- **Server-Side Signature & Verification:** Secure endpoint verifying payment completion before triggering order dispatch.

06 DELHIVERY LOGISTICS & AUTOMATED SHIPPING API

FULL COURIER AUTOMATION (/API/DELHIVERY/*, ORDERTRACKERMODAL.TSX)

Automated AWB & Live
Tracking

- **Live 6-Digit PIN Code Serviceability Engine:** Real-time validation against Delhivery database with auto-fill for City and State upon entering PIN code.
- **Instant Auto-Shipment Dispatch:** Automated AWB (Waybill) creation directly upon verified payment.
- **Embedded Live Tracking Modal:** 4-stage visual timeline (*Manifested* → *In Transit* → *Out for Delivery* → *Delivered*) directly on website.

07 CLOUD DATABASE & CUSTOMER ORDER HISTORY

CLOUD FIRESTORE PERSISTENCE (LIB/ORDERS.TS, ORDERHISTORYMODAL.TSX)

Real-Time Data Layer

- **Order Storage:** Complete order snapshots, customer info, itemized SKUs, payment gateway IDs, and AWB tracking numbers saved in Firestore.
- **Customer Order Portal:** Logged-in users can review past purchases, download itemized breakdown receipts, and check real-time courier statuses.

08 MULTI-CHANNEL NOTIFICATION & WHATSAPP SYNC

AUTOMATED CUSTOMER COMMUNICATIONS (/API/SEND-CONFIRMATION)

Resend HTML Email + WhatsApp

- **Branded Transactional HTML Emails:** Instant invoice emails dispatched via Resend API with itemized table, total amount, and direct tracking buttons.
- **WhatsApp Order Sync:** Instant dynamic WhatsApp notification link generated for immediate client updates.
- **Floating WhatsApp Support Widget (WhatsAppWidget.tsx):** Always-available customer service trigger.

09 ADVANCED SEO, SOCIAL MEDIA & STRUCTURED DATA

GOOGLE RICH RESULTS & RANK READY ARCHITECTURE (APP/PAGE.TSX)

JSON-LD Schema Markup

- **JSON-LD Structured Data:** Organization schema, WebSite search action schema, and Product/ItemList schemas with pricing, ratings, and shipping details.
- **OpenGraph & Twitter Card Protocols:** High-resolution social previews for WhatsApp, Instagram, and Twitter links.

10 LEGAL & COMPLIANCE INFRASTRUCTURE

REGULATORY COMPLIANCE PAGES

SSR Pages + Modal Views

- Dedicated regulatory compliant pages and interactive modals for **Privacy Policy**, **Refund Policy**, **Shipping Policy**, and **Terms & Conditions**.

11 COMMERCIAL VALUE & DELIVERABLES QUOTATION

Module / Deliverable Scope	Key Architectural Value	Market Value
01. Custom Headless Next.js 16 UI/UX & Animations	Custom Canvas 2D/3D Particle Engine, Web Audio Synthesizer, Sub-1s page speeds	₹35,000
02. Cashfree Payment Gateway Integration	Server-to-server security, in-app modal checkout, multi-channel UPI/Cards	₹15,000
03. Delhivery Logistics & Automated Shipping API	Live PIN code validation, automated AWB generation, real-time tracking modal	₹22,000
04. Cloud Database & Authentication (Firebase)	Google 1-Tap Auth, Guest checkout, Firestore user profile & order history portal	₹18,000
05. Multi-Channel Notification Engine	Automated Resend transactional HTML emails + instant WhatsApp order sync	₹10,000
06. Advanced SEO & Google Schema Data	Full JSON-LD structured schemas for Products, Offers, Ratings & Organization	₹10,000
07. Dynamic Coupon Engine & Cart Architecture	Percentage/Flat coupon validation with cart threshold criteria & real-time drawer	₹8,000
TOTAL DELIVERABLE VALUATION (CUSTOM ENTERPRISE STACK)		₹1,18,000
Platform Type	Monthly App Fees	Fulfillment Mode
Headless D2C	₹0 / Month	100% Automated
Stage & Steel — Athlete-Grade Sports Nutrition E-Commerce System		
Confidential & Proprietary Deliverables Document		